

Skills Summary

Expanded & Factored Forms

Use this reference while completing your worksheet or when studying.

Skill 1 — Expanding (distribute)

Rule: $a(b + c) = ab + ac$ — multiply the outside factor by every term inside. Expanded form has *no parentheses*.

Example: Expand $2(4x + 3)$

Step 1. To remove the parentheses, distribute the outside factor to each term inside, one product at a time.

Step 2. $2 \cdot 4x = 8x$ and $2 \cdot 3 = 6$.

$$\Rightarrow 2(4x + 3) = 8x + 6$$

Try it: Expand $5(3x + 2)$

check: $15x + 10$

Skill 2 — Combining like terms

Rule: like terms share the same variable part; add or subtract their coefficients and keep the variable part. Constants combine only with constants.

Example: Simplify $7x + 8 - 3x - 2$

Step 1. Group the terms by kind before combining: the x -terms together, the plain numbers together.

Step 2. x -terms: $7x - 3x = 4x$. Constants: $8 - 2 = 6$.

$$\Rightarrow 7x + 8 - 3x - 2 = 4x + 6$$

Try it: Simplify $10x + 1 - 6x + 4$

check: $4x + 5$

Skill 3 — Factoring out the GCF

Rule: factoring undoes expanding: find the greatest common factor of the terms and pull it outside: $ab + ac = a(b + c)$. Check by distributing back.

Example: Factor $15x + 20$

Step 1. Find the largest number dividing both coefficients; it becomes the outside factor and the quotients stay inside.

Step 2. The GCF of 15 and 20 is 5: $15x = 5 \cdot 3x$ and $20 = 5 \cdot 4$.

$$\Rightarrow 15x + 20 = 5(3x + 4)$$

Try it: Factor $8x + 12$

check: $4(2x + 3)$

Skill 4 — Mixed rewrites: expand, then combine

Rule: when an expression mixes a product with loose terms, expand the product *first*, then combine like terms. Work one rewrite at a time.

Example: Simplify $3(2x + 1) + 4x$

Step 1. The parentheses block the like terms from meeting, so expand the product first and only then combine.

Step 2. $3(2x + 1) = 6x + 3$; then $6x + 3 + 4x = 10x + 3$.

$$\Rightarrow 3(2x + 1) + 4x = \mathbf{10x + 3}$$

Try it: Simplify $2(3x + 4) - 5x$

8 + x :xəəp